

Spotlight

Where you start could determine where you end up

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A significant subset of the eukaryotic proteome must localize to multiple cellular compartments, necessitating the emergence of mechanisms to produce protein isoforms with distinct targeting sequences. Ly et al. reveal that alternative start-codon selection during translation initiation is a pervasive mechanism for dual localization of proteins by generating N-terminal isoforms.

Eukaryotes have evolved distinct subcellular compartments to carry out specialized functions, employing a dedicated set of proteins that are often synthesized by cytosolic ribosomes. These proteins typically have targeting sequences that are used by translocation/import machinery to localize them to the correct compartment. Yet, in addition to these compartment-specific functions, cellular compartments must also perform functions that overlap with other compartments. For instance, to replicate its genome, the mitochondrion often employs enzymes that are also used to replicate the nuclear genome. Similarly, maturation of mitochondrial tRNAs uses a set of enzymes that overlap with those used for cytosolic tRNAs. As a result, these proteins must localize to distinct compartments. To some extent, evolution appears to have solved this problem through gene duplication, during which one paralog acquired a compartment-targeting sequence [1]. Additional mechanisms include alternative transcription initiation and splicing, which generate different mRNA isoforms [2,3]. Notably, recent

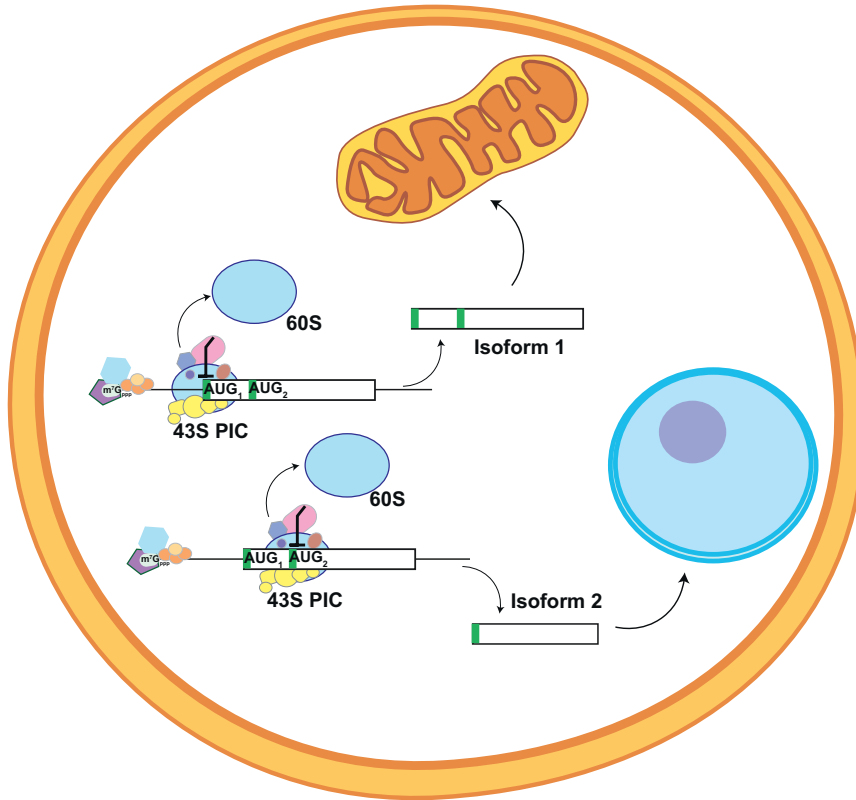
evidence suggests that translation-based mechanisms also contribute to dual localization of proteins. For instance, in yeast, translation of the mRNA encoding the DNA helicase Pif1 produces two protein isoforms, with one isoform targeted to the nucleus and the other to the mitochondria [4]. Similarly, in *Arabidopsis thaliana*, the mRNA-encoding POLy2 produces two protein products that localize to the nucleus and chloroplast [5]. These events take advantage of alternative start-codon selection by the scanning small ribosomal subunit during translation initiation.

During alternative start-codon selection, the mRNA must possess at least one additional initiation codon in addition to the ‘canonical’ one. If the alternative initiation codon is in frame with the canonical one, protein isoforms with varying lengths are produced. Depending on its location relative to the annotated initiation codon, an in-frame alternative start codon can produce N-terminally truncated or N-terminally extended products. To produce distinct isoforms, translation initiation involves leaky ribosomal scanning, during which the 43S pre-initiation complex skips the upstream start codon and instead initiates translation downstream. Leaky scanning by the small subunit is influenced by factors such as the strength of the Kozak sequence surrounding a start codon and the presence of stem-loop structures upstream of the start codon [6].

Although leaky scanning has been shown to regulate dual localization of some proteins in yeast, plants, and green algae [4,5,7], the pervasiveness of this mechanism in mammals and the potential link between its dysregulation and human disease have not been fully appreciated. To this end, in a recent study, Ly et al. (2024) used ribosome profiling to map transcriptome-wide translation start sites. They found that, on average, each open reading frame in HeLa cells harbors two initiation codons [8]. In the new issue of

Molecular Cell [9], they elaborated on these observations and asked whether alternative start-codon selection is used by human cells to produce different protein isoforms for localization in distinct subcellular compartments. Of the ~7500 gene transcripts in their ribosome-profiling data, more than 2500 transcripts contain in-frame start codons that are predicted to produce isoforms with N-terminal extensions and truncations. Since most protein-targeting sequences are found at their N terminus, the authors hypothesized that these alternative N termini may impact protein localization. Indeed, the subcellular localization prediction algorithm DeepLoc2.1 predicted that more than 25 and 10% of N-terminally truncated and extended isoforms, respectively, would display differential protein localization. Out of 40 proteins that were identified by DeepLoc2.1, 27 exhibited differential protein localization when tested using green fluorescent protein (GFP) tagged constructs. The authors decided to focus on mitochondria-localized proteins and asked how the N-terminal extensions or truncations impact targeting. Remarkably, for some dually localized proteins, alternative translation initiation appears conserved across Eukarya. *A priori*, the most logical model that could explain dual localization of a protein is that extensions and truncations lead to inclusion or exclusion of a targeting sequence (Figure 1). Notably, however, the authors identified a previously unappreciated and pervasive mechanism whereby an additional N-terminal sequence masks the recognition of the downstream targeting sequence. For some proteins, N-terminal extension of just seven amino acids was sufficient to prevent the protein from localizing to the mitochondria.

Out of the proteins that the authors identified to exhibit differential localization, those involved in tRNA maturation and aminoacylation in multiple compartments make sense. Yet, for AURKAIP1/MRPS38, which is predicted to exclusively



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Figure 1. Schematic showing how dual protein localization can be achieved through alternative start-codon selection. Due to leaky scanning, a 43S pre-initiation complex (PIC) often skips the first start codon (AUG₁) and instead initiates on an in-frame downstream start codon (AUG₂). Initiation at AUG₁ results in translation of the extended isoform 1, which in this case is imported to the mitochondria. Initiation at AUG₂ results in translation of the truncated isoform 2, which in this case is exported to the nucleus.

function in the mitochondria as a component of the mitoribosome, the localization of one of its isoforms to the nucleolus at first glance did not make sense unless it has a previously unrecognized role there. Indeed, Ly *et al.* (2025) went on to show that the extended protein interacts with ribosome biogenesis factors, suggesting that the protein has a role in cytosolic ribosome assembly. It will be interesting to investigate how this mitochondrial ribosomal protein contributes to cytosolic ribosome assembly. More importantly, what about the other ~200 proteins that exhibit dual localization? Do any of their isoforms carry out functions that are different from those they perform in their 'assigned' compartments?

Having established that N-terminal isoform-mediated dual localization plays important roles in maintaining the function of a set of proteins in the mitochondria and elsewhere, Ly *et al.* (2025) next sought to explore whether certain disease-derived mutations could impact dual localization. In particular, the authors analyzed the ClinVar database and found that, of the ~3 000 000 variants analyzed, ~25 000 were predicted to alter N-terminal isoform production of ~1000 proteins. More than 250 of the genes encoding these proteins harbored nonsense and frameshift mutations downstream of the annotated start codon, but upstream of the alternative start codon. As expected, these mutations were

previously considered as a loss of function. However, in light of the analysis conducted by Ly *et al.*, these mutations are predicted to abrogate the production of the annotated protein, but not the truncated form. Consistent with these ideas, analysis of a subset of these mutations using GFP-tagged constructs revealed that they alter the production of the mitochondrial isoforms, but not the cytosolic and nuclear ones. In contrast to nonsense and frameshift mutations, missense mutations alter the sequence of the targeting sequence and hence may inhibit recognition by the mitochondrial import machinery. Indeed, for the PNPO gene, an R6W mutation, which has been associated with pyridoxal phosphate-responsive seizures, reduced import to the mitochondria without affecting the production of the downstream protein product. Notably, since these variants affect the production of only one protein isoform, their disease pathology should be distinct from those that lead to the complete loss of function. Consistent with these ideas, patients with sideroblastic anemia, B cell immunodeficiency, periodic fevers, and developmental delays (SIFD) that carried mutations in *TRNT1*, which were only predicted to selectively impact a single *TRNT1* isoform, did not exhibit typical SIFD symptoms. Overall, the authors' data suggested that when assessing the contribution of a mutation to a disease, it is important to consider its effect on the production and/or targeting of protein isoforms.

In summary, Ly *et al.* (2025) revealed that alternative start-codon selection is a widespread and conserved mechanism that evolved to generate dual-localized mitochondrial and cytosolic/nuclear protein isoforms. They also went on to show that disruption of the process can lead to unique disease phenotypes that result from compartment-specific loss of function. These findings add important insights into the role of translational control in producing different protein isoforms that

localize to distinct cellular compartments. Additionally, the study provides new avenues to explore. Arguably, one of the most important questions to address is whether the process can be regulated to adjust the ratio of the two isoforms in response to mitochondrial demand, for example. Interestingly, in an earlier study, the authors showed that the frequency of alternative start-codon selection is regulated during mitosis through the release of nuclear eIF1 [8]. Does a similar mechanism exist to regulate dual localization through alternative start-codon selection? Also, what are the advantages of this mechanism relative to others? For example, gene duplication to encode paralogs with extended or truncated N termini is a more energetically expensive process, but at the same time provides more control over isoform expression regulation. Equally important is the mechanism by which the mRNA structure and sequence regulate leaky scanning. For MRPS38, the authors showed that a near-cognate

codon and a stem loop are important determinants. This, in turn, begs the question of whether mRNA modification and/or helicases regulate alternative start-codon selection. Finally, as the authors alluded to, it will be interesting to explore the role of this mechanism in localizing proteins to compartments other than the mitochondria.

Acknowledgments

Work in the Zaher Lab is supported by the NIH Grants R35GM158294, R01GM112641, and R01GM141474 to H.S.Z.

Declaration of interests

The authors declare no competing interests.

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<https://doi.org/10.1016/j.tcb.2025.12.003>

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